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PHASE-FIELD SIMULATIONS OF NON-ISOTHERMAL BINARY ALLOY SOLIDIFICATION

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Abstract—A phase-field method for two-dimensional simulations of binary alloy solidification is studied. Phase-field equations that involve both temperature and solute redistribution are formulated. The equations are solved using the finite element method with triangular elements on unstructured meshes, which are adapted to the solution. Dendritic growth into a supersaturated melt is simulated for two temperature regimes: (a) the temperature is prescribed on the boundary of the computational domain; and (b) the heat is extracted through the domain boundary at a constant rate. In the former regime the solute redistribution is compared with the one given by an isothermal model. In the latter case the influence of the size of the computational domain and of the heat extraction rate on dendritic structure is investigated. It is shown that at high cooling rates the supersaturation is replaced by thermal undercooling as the driving force for growth. *© 2001 Acta Materialia Inc. Published by Elsevier Science Ltd. All rights reserved.*

Keywords: Phase-field; Kinetics; Diffusion

1. INTRODUCTION

Over the last 10 years the phase-field method has been extensively used for simulations of dendritic growth. In the phase-field method a new variable $\phi(x, y, t)$ is introduced to indicate the physical state of the system at each point. ϕ takes on constant values in solid and liquid and changes steeply but smoothly over a thin transition layer that plays the role of the classical sharp interface. The governing equation coupled with modified transport equations are applied in all of space without distinguishing between the phases. This permits simulations of growing morphologies without explicitly tracking the phase boundaries.

Solidification of a binary mixture has been studied by Caginalp *et al*. [5, 8]. It is shown that the phase-field equations reduce to the traditional sharp interface models in the limit when the thickness of the interfacial region δ vanishes. At the same time computations demonstrate that the phase-field methods produce an interface close to the sharp interface problem even for relatively large δ.

Despite the great success in predicting qualitatively realistic microstructures, the phase-field method has only been applied to very simple systems. Typically heat flow during solidification of pure substances or isothermal diffusion in binary alloys obeying ideal solution thermodynamics have been studied. However, from a technological point of view, it would be valuable to perform simulations on real alloys, which are usually multicomponent and have complex thermodynamic interactions.

Warren and Boettinger [1] derived a phase-field model for isothermal solidification of a binary alloy, applying constant diffusivities within the solid and liquid phases, and performed two-dimensional simulations of dendritic growth into a highly supersaturated liquid. This model has been explored in several papers, for example, [9–12]. A desirable extension of the model is to study the effect of heat flow due to release of latent heat. However, the numerical implementation is not trivial since the temperature and solute evolution occurs on completely different time-scales. A simplified approach was proposed in Ref. [4], where the spatial variation of the temperature is neglected and the heat equation is replaced by a heat balance of an imposed heat extraction rate and the latent heat release rate.

The present report is part of a project where the ultimate goal is to apply the phase-field method to simulate processing of real alloys. As a first step, the phase-field formulation [1] is applied, with the major difference being that simultaneous heat flow and dif-

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fusion are taken into account. The paper is organized as follows: Section 1 gives an overview of the model derivation; numerical aspects are presented in Section 2; in the next two sections results of truly non-isothermal calculations of dendritic growth for two temperature regimes are presented and discussed.

2. MATHEMATICAL MODEL

In this section the phase-field model given in Ref. [1] is considered with modifications accounting for temporal and spatial evolution of the temperature field. The formulation is based on an entropy functional

$$S = \int_{\Omega} \left(s(\phi, x_{\mathrm{B}}, e) - \frac{\epsilon^2}{2} |\nabla \phi|^2 \right) \mathrm{d}\Omega \tag{1}$$

where the thermodynamic entropy density s is a function of the phase-field variable ϕ varying smoothly between 0 in the solid and 1 in the liquid, the mole fraction x_{B} of a solute B in solvent A and the internal energy density e, and Ω is a spatial region occupied by a mixture.

Anisotropy is included in the system because the phase change kinetics depends upon the orientation of the interface

$$\epsilon = \bar{\epsilon}\eta = \bar{\epsilon}(1 + \gamma \cos k\beta) \tag{2}$$

where $\bar{\epsilon}$ is related to the surface energy σ and interface thickness, γ is the magnitude of anisotropy in the surface energy, k specifies the mode number and the expression $\beta = \arctan(\phi_y/\phi_x)$ gives an approximation of the angle between the interface normal and the orientation of the crystal lattice.

The evolution of the non-conserved phase-field variable is governed by

$$\dot{\phi} = M_{\phi} \frac{\delta S}{\delta \phi} \tag{3}$$

where M_{ϕ} is related to the interfacial mobility. The evolution of x_{B} and e is governed by the normal conservation laws

$$\frac{\dot{x}_{\mathrm{B}}}{V_{\mathrm{m}}} = -\nabla \cdot \mathbf{J}_{\mathrm{B}} \tag{4}$$

$$\dot{e} = -\nabla \cdot \mathbf{J}_e \tag{5}$$

where V_{m} is the molar volume. The diffusional flux $\mathbf{J}_{\mathrm{B}}$ and the heat flux $\mathbf{J}_e$ are given by the linear laws of irreversible thermodynamics

$$\mathbf{J}_B = L^{\mathrm{A}}_{\mathrm{BB}} \nabla \frac{\delta S}{\delta x_{\mathrm{B}}} + L^{\mathrm{A}}_{\mathrm{B}e} \nabla \frac{\delta S}{\delta e} \tag{6}$$

$$\mathbf{J}_e = L^{\mathrm{A}}_{e\mathrm{B}} \nabla \frac{\delta S}{\delta x_{\mathrm{B}}} + L^{\mathrm{A}}_{ee} \nabla \frac{\delta S}{\delta e} \tag{7}$$

$L^{\mathrm{A}}_{\mathrm{BB}}$ and L^{A}_{ee} are related to the inter-diffusional mobility of B and A and the heat conduction, respectively. The coefficient $L^{\mathrm{A}}_{\mathrm{B}e} = L^{\mathrm{A}}_{e\mathrm{B}}$ describes the cross effects between heat flow and diffusion and will be neglected. The second law requires M_{ϕ}, $L^{\mathrm{A}}_{\mathrm{BB}}$ and L^{A}_{ee} to be positive. It should be emphasized that the gradients $\nabla(\delta S/\delta x_{\mathrm{B}})$ and $\nabla(\delta S/\delta e)$ are to be evaluated isothermally and under fixed composition, respectively. The variational derivatives in equations (6) and (7) are given by

$$\frac{\delta S}{\delta x_{\mathrm{B}}} = \frac{\partial s}{\partial x_{\mathrm{B}}} = -\frac{\mu_{\mathrm{B}} - \mu_{\mathrm{A}}}{T V_{\mathrm{m}}} \tag{8}$$

$$\frac{\delta S}{\delta e} = \frac{\partial s}{\partial e} = \frac{1}{T} \tag{9}$$

where T is the temperature and the quantity on the right-hand side of equation (8) is known as the inter-diffusion potential in binary substitutional alloys. The chemical potentials μ_{A} and μ_{B} under the assumption of an ideal mixture have the following form

$$\mu_{\mathrm{A}} = {}^{\circ}\mu_{\mathrm{A}}(\phi, T) + RT \ln(1 - x_{\mathrm{B}}) \tag{10}$$

$$\mu_{\mathrm{B}} = {}^{\circ}\mu_{\mathrm{B}}(\phi, T) + RT \ln x_{\mathrm{B}} \tag{11}$$

The expressions of the molar Gibbs energy for pure materials are presented as in Ref. [7]

$$\frac{{}^{\circ}\mu_{\mathrm{A}}}{V_{\mathrm{m}}} = W_{\mathrm{A}} g(\phi) T + [e^{s}_{\mathrm{A}}(T^{\mathrm{A}}_{\mathrm{m}}) - c_{\mathrm{A}} T^{\mathrm{A}}_{\mathrm{m}} + p(\phi) \Delta H_{\mathrm{A}}] \left(1 - \frac{T}{T^{\mathrm{A}}_{\mathrm{m}}}\right) - c_{\mathrm{A}} T \ln \frac{T}{T^{\mathrm{A}}_{\mathrm{m}}} \tag{12}$$

$$\frac{{}^{\circ}\mu_{\mathrm{B}}}{V_{\mathrm{m}}} = W_{\mathrm{B}} g(\phi) T + [e^{s}_{\mathrm{B}}(T^{\mathrm{B}}_{\mathrm{m}}) - c_{\mathrm{B}} T^{\mathrm{B}}_{\mathrm{m}} + p(\phi) \Delta H_{\mathrm{B}}] \left(1 - \frac{T}{T^{\mathrm{B}}_{\mathrm{m}}}\right) - c_{\mathrm{B}} T \ln \frac{T}{T^{\mathrm{B}}_{\mathrm{m}}} \tag{13}$$

where $g(\phi) = \phi^2(1-\phi)^2$ and W_{A}, W_{B} are constants. $e^{s}_{\mathrm{A}}(T^{\mathrm{A}}_{\mathrm{m}})$ and $e^{s}_{\mathrm{B}}(T^{\mathrm{B}}_{\mathrm{m}})$ are the energy densities of pure

solid A and B at their melting points T_m^A and T_m^B, respectively. ΔH_A and ΔH_B are the heats of fusion per volume, c_A and c_B are the heat capacities and R is the gas constant. $p(\phi)$ is a smoothing function, chosen such that $p'(\phi) = 30g(\phi)$.

The phase-field equation (3) in the present model is used as derived in Ref. [1]

$$\dot{\phi} = M_\phi \epsilon^2 \left[\nabla\cdot(\eta^2\nabla\phi) - \frac{\partial}{\partial x}\left(\eta\eta_\beta'\frac{\partial\phi}{\partial y}\right) + \frac{\partial}{\partial y}\left(\eta\eta_\beta'\frac{\partial\phi}{\partial x}\right)\right] - M_\phi((1-x_B)H_A + x_B H_B) \tag{14}$$

where

$$H_A(\phi, T) = W_A g'(\phi) + 30g(\phi)\Delta H_A\left(\frac{1}{T} - \frac{1}{T_m^A}\right) \tag{15}$$

$$H_B(\phi, T) = W_B g'(\phi) + 30g(\phi)\Delta H_B\left(\frac{1}{T} - \frac{1}{T_m^B}\right)$$

The diffusion equation is now obtained by combining equations (4), (6), (8) and (10)–(13). However, as pointed out earlier, the gradient $\nabla(\delta S/\delta x_B)$ must be evaluated isothermally, therefore the diffusional flux may be written

$$\mathbf{J}_B = L_{BB}^A\left[-\frac{R}{x_B(1-x_B)}\nabla x_B + (H_A(\phi, T) - H_B(\phi, T))\nabla\phi\right] \tag{16}$$

Comparison of equation (16) with the normal Fick's law of diffusion gives that

$$L_{BB}^A = D\frac{x_B(1-x_B)}{RV_m} \tag{17}$$

where D is the normal Fickian coefficient of interdiffusion of A and B. In this case the so-called thermodynamic factor is unity because the ideal solution is assumed. The final diffusion equation is then obtained by combining equations (4), (16) and (17) and takes the form

$$\dot{x}_B = \nabla\cdot D\left[\nabla x_B + \frac{V_m}{R}x_B(1-x_B)(H_B(\phi, T) - H_A(\phi, T))\nabla\phi\right] \tag{18}$$

The diffusion coefficient is postulated as a function of the phase-field variable

$$D = D_S + p(\phi)(D_L - D_S) \tag{19}$$

where D_S, D_L are the classical diffusion coefficients in the solid and liquid, respectively.

To complete the derivation of the model, the internal energy density is postulated according to Ref. [5]

$$e = (1-x_B)e_A + x_B e_B \tag{20}$$

The internal energy densities of pure materials under the assumption of equal solid and liquid heat capacities are

$$e_A = e_A^s(T_m^A) + c_A(T - T_m^A) + p(\phi)\Delta H_A \tag{21}$$

$$e_B = e_B^s(T_m^B) + c_B(T - T_m^B) + p(\phi)\Delta H_B$$

Inserting equation (20) into equation (5) with $L_{ee}^A = KT^2$ implies that the heat equation has the form

$$\tilde{c}\dot{T} + 30g(\phi)\Delta\tilde{H}\dot{\phi} + N\dot{x}_B = \nabla\cdot K\nabla T \tag{22}$$

where the following formulae are introduced

$$\tilde{c} = (1-x_B)c_A + x_B c_B \tag{23}$$

$$\Delta\tilde{H} = (1-x_B)\Delta H_A + x_B\Delta H_B \tag{24}$$

and $N = \partial e/\partial x_B$. Similar to the heat capacity and the latent heat of fusion of the mixture the thermal conductivity of the mixture is approximated by a weighted sum of conductivities of the pure materials

$$K = (1-x_B)K_A + x_B K_B \tag{25}$$

Again, equal solid and liquid thermal conductivities of both materials are assumed. Following Ref. [8], the heat equation is simplified by dropping the $\dot{x}_B$ term

$$\tilde{c}\dot{T} + 30g(\phi)\Delta\tilde{H}\dot{\phi} = \nabla\cdot K\nabla T \tag{26}$$

A similar heat equation was derived in Ref. [12] for the case of a dilute alloy.

3. NUMERICAL ISSUES

For convenience, the governing equations (14), (18) and (26) are transformed into dimensionless form. Length and time have been scaled with a reference length $l = 0.94\delta$ and the diffusion time l^2/D_L, respectively. The non-dimensional temperature is

defined as $\theta = (T-T^*)/\Delta T$, where T^* is the temperature of the liquidus associated with an initial solute composition.

In the present phase-field formulation the interface thickness is treated as an input parameter. From a physical point of view, it is desirable to take δ close to the capillary length $l_0=7.1\times10^{-10}$ m. However, this requires extremely dense grids and time-consuming computations. To validate the choice of δ, the steady growth of a dendrite tip was investigated for $\delta=4.9\times10^{-8}$, 1.7×10^{-8}, 4.5×10^{-9} and 2.2×10^{-9} m. The same rate of convergence of the liquid concentration at the interface x_B towards the sharp interface theory prediction x_B^{SI} was obtained here, as that reported in Ref [1]. The dendrite tip speed for these values of δ is 1.2, 1.0, 0.94, and 0.93 cm/s, respectively. With $\delta=4.9\times10^{-8}$ m, as chosen for the present calculations, the tip speed thus varies $<30\%$ when the interface thickness is reduced by a factor of 20, almost down to the nominal value. This indicates that the phase-field method produces an interface reasonably close to the sharp interface problem even for relatively large δ.

Following Ref. [1] a Ni–Cu alloy is chosen for the simulations. All the phase-field parameters are related to the physical properties of the alloy and given in Ref. [1]. The physical data used in the calculations are presented in Table 1. The magnitude of anisotropy γ is 0.04 and $k=4$ employs four-fold symmetry, which allows one to reduce computational costs by performing calculations only in one-fourth of the domain.

The whole region initially contains supersaturated (0.86) and undercooled ($\Delta T=20.5$K) melt, specifically for the alloy composition $x_B^o = 0.4083$ and $T=1574$K, except for a small nuclei of a circular shape with non-dimensional radius $r_0=2$, placed in the centre of the domain. The initial distribution of ϕ is based on the solution of one-dimensional phase-field equation for isothermal coexistence of liquid and solid at a planar interface of a pure metal

$$\phi(x, y, 0) = \frac{1}{2}\left[1 + \tanh\left(\frac{\sqrt{x^2 + y^2}-r_0}{2\sqrt{2}}\right)\right] \quad (27)$$

Zero Neumann boundary conditions for x_B and ϕ are imposed at the boundaries. The boundary conditions for temperature will be defined later.

It was shown in Ref. [1] that stochastic noise introduced into the phase-field model causes fluctuations at the solid/liquid interface, that leads to the development of a dendritic structure. According to Refs. [13,14] the physical origin of the noise is stochastic forces appearing in the system due to thermodynamic fluctuations near the dendrite tip. The noise is included as proposed in Ref. [1] by modifying the phase-field equation

$$\dot{\phi}\rightarrow\dot{\phi}-M_\phi\alpha r 16 g(\phi)((1-x_B)H_A + x_B H_B) \quad (28)$$

where r is a random number distributed uniformly between -1 and $+1$, a new number is generated for every point of the grid, at each time-step. α is an amplitude of the fluctuations taken as 0.4.

A specific feature of binary alloy solidification is that the changes of x_B and ϕ are highly localized over the solid/liquid interface. The width of the interface is much smaller than the other length scales, which makes the use of an adaptive unstructured mesh beneficial. Initially the computational domain is spatially discretized with large triangular elements and then the interface region gets the highest resolution. As the interface evolves during the computation the mesh is adaptively changed by splitting or merging the elements according to an error-function

$$\mathrm{Err}\approx\int_{\Gamma_{el}}(\nabla\phi + \nabla\theta + 25\phi\nabla x_B)\cdot\bar{n}\,d\Gamma + 100\int_{\Omega_{el}}\phi^2(1-\phi)^2 d\Omega \quad (29)$$

where Ω_{el} is the area of an element and Γ_{el} is its boundary with normal $\bar{n}$. This technique provides accurate results with a minimum number of elements. For more details the reader is referred to Ref. [6].

The system of non-dimensional equations is transformed into a discrete problem by the use of Galerkin formulation of the finite element method, with linear basis functions. The time derivatives are discretized by a first order finite difference approximation, the diffusion terms are made implicit and when it is possible the other terms are also chosen implicitly. Although the numerical scheme is quite stable, the time-step is chosen significantly below the stability limit to satisfy an accuracy requirement. A Fortran code was developed by means of FemLego [2,3] which performs automatic code generation from a high-level Maple specification.

The numerical scheme was verified on the isothermal model, equations (14) and (18). A square computational domain of non-dimensional size 750×750 is used in the simulations. Minimum and maximum mesh resolution was defined as a compromise between number of gridpoints and an implicit numerical noise produced by the model. With 0.625 and 10 no secondary sidebranches are developed

Table 1. Physical data for Ni, Cu

	Nickel(A)	Copper (B)
T_m (K)	1728	1358
ΔH (J/m^3)	2350×10^6	1728×10^6
σ (J/m^2)	0.37	0.29
D_L (m^2/s)	10^{-9}	10^{-9}
D_S (m^2/s)	10^{-13}	10^{-13}
c [J/(Km^3)]	5.42×10^6	3.96×10^6
K [J/(Ksm)]	84.0	200.0

when $\alpha=0$. The steady-state growth velocity of the dendrite tip was chosen as a reliable quantity to define time-step. It was found that with $\Delta t=0.2$ the numerical scheme gives an error of the tip speed ca 3%. Solute redistribution obtained for isothermal calculations is given in Fig. 1.

4. TEMPERATURE FIXED ON THE BOUNDARY

To complete the model describing cooling of a melt with growing nuclei, thermal boundary conditions are to be specified. Different types of boundary conditions determine different thermal regimes affecting the temperature variation in space. In this section

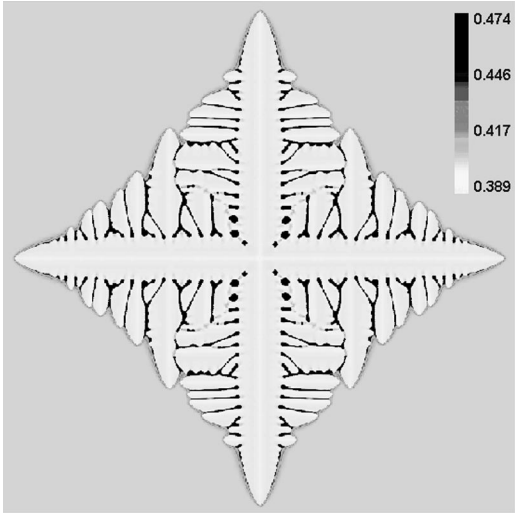


Fig. 1. Concentration field, isothermal model. Time is 2.75 ms.

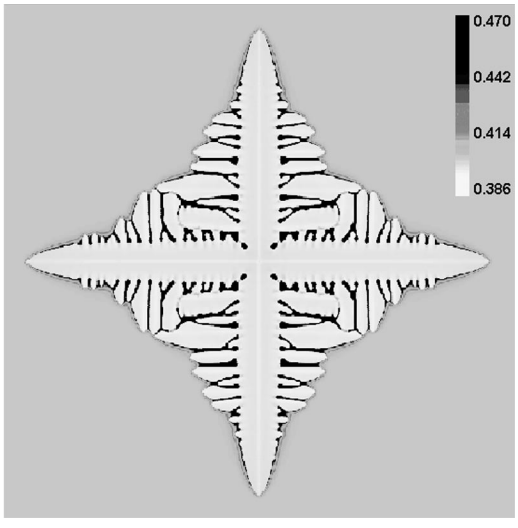


Fig. 2. Concentration field given by the non-isothermal model with temperature fixed on the boundary. All the physical and phase-field parameters are the same as in isothermal calculations. Time is 2.75 ms. The color scheme uses yellow–red palette in grey scale for low and high concentration values, respectively.

equations (14), (18) and (26) are solved with $T=1574$K as initial and boundary conditions for the heat equation. This is done in order to fit the non-isothermal temperature regime to the isothermal case. The results of this simulation are shown in Figs 2 and 3.

As one would expect, the temperature does not vary much, neither in time nor in space, for example, the spatial temperature difference does not exceed 1.6K all the time (Fig. 4). However, comparison of the solute patterns (Figs 1 and 2) shows that the composition behaviour is very sensitive to even small changes in the temperature field. Increased melt temperature reduces sources of instability, which leads to less developed structure of the non-isothermal dendrite. The length of the primary arms is about 6% less than the corresponding one calculated under isothermal conditions. The solute diffusion length is larger compared to the isothermal growth, as well as spacing of interdendritic liquid pockets. The value of the composition at the solid/liquid interface is decreased since the operating point in the phase diagram has moved to the left due to the increased temperature.

Figure 3 demonstrates spatial redistribution of the temperature field with imposed interface location at time 2.75 ms. As predicted by the Gibbs–Thomson condition, the tips are the coldest parts of the dendrite due to the large curvature here. The location of the hottest point during the crystal growth varies and in general corresponds to one of the tips of those sidebranches which grow towards each other and form a closed liquid pocket. Melt in this pocket is of highest temperature due to release of latent heat by these growing sidebranches. The less sharp the sidebranch tip, the higher temperature of the sidebranch.

The maximal value of the system temperature as a function of time is given in Fig. 4. The curve is slightly oscillating because the temperature is taken at different grid points wherever the maximum value occurs. The local peaks occur when two or more sidebranches growing towards each other merge and stop growing and consequently to produce the latent heat. Then another sidebranch surrounded by hot melt starts to release more latent heat than the others and becomes the hottest place in the system. Fig. 5 shows the area of the dendrite tip with imposed isotherms, which are refracted over the interface. This reflects the change of the temperature gradient owing to the release of latent heat.

The results presented in this section indicate clearly that for a supersaturated binary alloy the development of dendritic patterns is governed mainly by solute diffusion, but the temperature variation alters significantly the morphology of the microstructure.

5. RECALESCENCE CALCULATIONS

This section represents the results obtained for simulations of dendritic solidification in the thermal

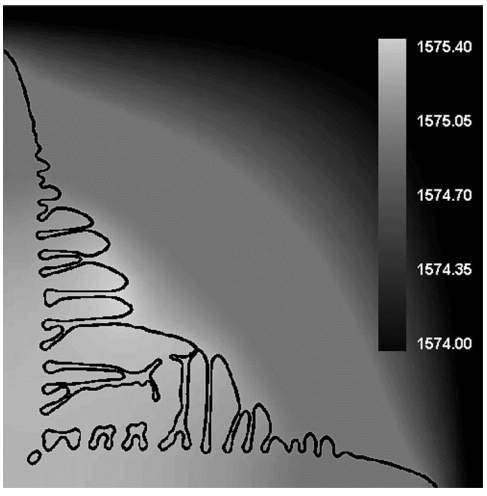


Fig. 3. Temperature field corresponding to the solute redistribution in Fig. 2. The lighter color, the higher the temperature. The solid/liquid interface is shown as a band where $0.1<\phi<0.9$. Time is 2.75 ms.

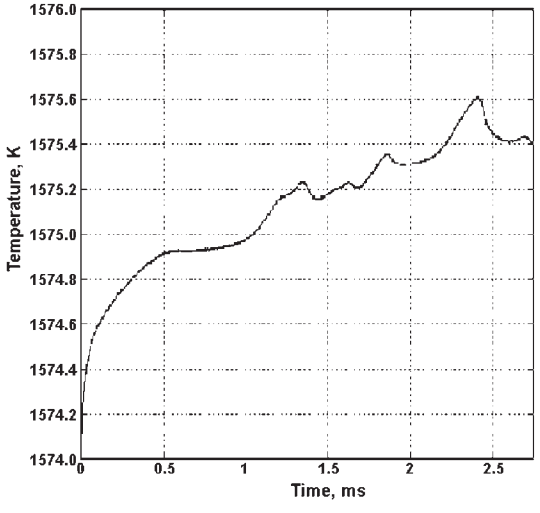


Fig. 4. Maximal temperature of the system vs time, minimal temperature $T=1574$K is kept on the boundary all the time.

regime when a heat flux is extracted from the domain. This regime is modeled in two ways: (a) the isothermal model is coupled with a heat balance [4] which neglects temperature variations in space

$$\frac{\mathrm{d}T}{\mathrm{d}t} = -\dot{T} + \frac{\Delta\tilde{H}}{\tilde{c}}\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{1}{S}\int_S \phi(x, y, t)d\mathrm{S}\right] \qquad (30)$$

where $\dot{T}$ is the cooling rate, S is the area of the domain, $\Delta\tilde{H}$ and $\tilde{c}$ are evaluated for the composition far away from the interface; (b) non-isothermal model with Neumann boundary conditions for the heat equation. The heat flux Q imposed on the outer boundary is related to $\dot{T}$ by $Q = \dfrac{\tilde{c}\dot{T}a}{4}$, where a is the side length of the computational domain.

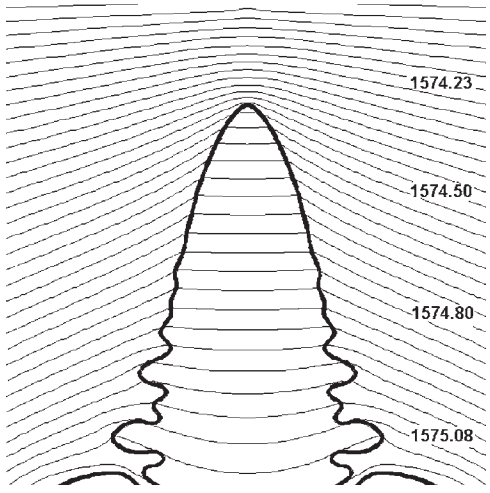


Fig. 5. Location of isotherms near the dendrite tip. The picture corresponds to the run in Fig. 3.

There is a lot of numerical and physical issues, which should be investigated for these types of simulations. In these calculations the effect of the cooling rate and the size of the domain is analyzed. The computations are performed in two square domains 6.9×10^{-5} and 2.28×10^{-5} m on a side (which will be referred to as "large" and "small" boxes), for three cooling rates: 2.1×10^{3}, 3.4×10^{4}, and 1.3×10^{5} K/s. In order to keep the same non-dimensional geometry, $\delta=1.62\times10^{-8}$ m is chosen for the "small" box.

Figure 6(a and b) shows the temperature–time history for all cooling rates, in the "big" and "small" box, respectively. Three curves represent each simulation: two dashed lines show the minimal and maximal value of the system temperature calculated by the non-isothermal model, the solid line is $T(t)$ obtained through the heat balance [equation (30)]. It is necessary to note that $T(t)$ calculated by the isothermal approach differs significantly (2K as the worst) from the short-term $T(t)$ dependence in Fig. 6 in Ref. [4]. $T(t)$ is very sensitive to the choice of numerical as well as physical parameters which were not unambiguously defined in Ref. [4].

In general, both the models give a similar $T(t)$ behaviour. Initially, at the highest cooling rates (3.4×10^{4} and 1.3×10^{5} K/s) the temperature falls down because the composition needs more time to be changed and to cause solidification. As a crystal starts to grow, the latent heat release increases the temperature, that is, recalescence occurs. For the lowest cooling rate, the initial growth of the nuclei is fast enough to overcome the imposed heat extraction rate and, hence, the temperature–time curve initially has a positive slope. $T(t)$ given by the isothermal model initially approximates an average temperature, but later the latent heat of fusion is released faster, and therefore, for all three cooling rates the time evolution is faster. As a result, the isothermal model gives an overestimated values of the temperature.

Comparison of the dashed curves in Fig. 6(a and

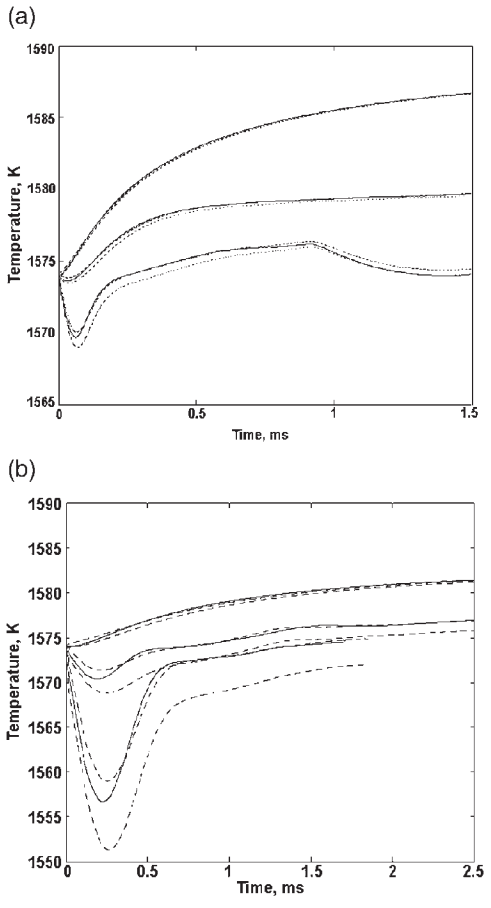


Fig. 6. Temperature–time dependence for the boxes of size 2.28×10^{-5} m (a) and 6.9×10^{-5} m (b). Results for the cooling rates 2.1×10^{3}, 3.4×10^{4}, and 1.3×10^{5} K/s are shown from top to bottom.

b) obtained for the same cooling rates shows the influence of the domain size. In outline, at the same time the temperature of the melt in the "large" box has a lower value than the temperature in the "small" domain. This feature can be explained by the fact that a larger box contains fewer nuclei per area, and thus, less latent heat is produced in the box. It is interesting to notice that a larger computational domain and higher heat extraction rate cause greater spatial variations of the temperature field. The maximum temperature alteration is ca 9° in the "large" box and 1.2° in the "small" one. The tendency of a slightly decreasing difference of maximum and minimum temperature as time goes is due to reducing fraction of melt. Therefore, the assumption made in Ref. [4] about absence of the temperature gradient in space can be accepted for a small computational domain with a low imposed heat extraction rate, but in other cases spatial variation of the temperature field should be taken into account.

The calculated growth morphologies vary significantly among the three cooling rates and the computational domains. Solute patterns and corresponding temperature fields are presented in Figs 7 and 8. Comparing these figures one should remember that the real side sizes of the boxes differ by a factor of three. At the lowest cooling rate [Figs 7(e) and 8(e)] the temperature rise reduces the melt undercooling which gives the least developed crystal. In this case the solidification is driven by solute change in both phases, solid and liquid. The higher the heat extraction rate, the greater the influence of the size of the computational domain on the crystal growth. While the dendritic morphology calculated in the "small" box [Fig. 7(c)] consists only of primary stalks with no perturbation on the interface, the corresponding morphology in the "large" box [Fig. 8(c)] has well developed secondary arms. It should be noted, that due to the high value of heat extraction rate the interface stays planar (which gives the crystal a diamond shape) before perturbations appear at the interface and secondary sidebranches start to develop.

The most intriguing morphology is presented in Fig. 8(a) and obtained for the highest cooling rate in the "large" domain. The morphology growth exhibits an almost circular shape with cells developing at the later times. The explanation is that for very high cooling rate, the solidification of a binary alloy is governed mainly by the heat transfer. When a melt freezes quickly, composition does not have time to be changed and to cause instability. The presence of solute trapping in this simulation is observed when the interface velocity V reaches its maximum value of 0.03 m/s, which corresponds to the largest undercooling of the melt. The partition ratio at the moment is $c_S/c_L = 0.97$, as opposed to an equilibrium value of 0.85.

The solute trapping effect in the phase-field models was studied in Ref. [15]. It was shown that for a planar interface the solute trapping occurs when the solute diffusion length ahead of the interface is comparable with the interface thickness $D_l/V \sim \delta$ where $D_l = D(\phi = 0.5)$. For the δ used in the calculations this gives $V = 0.01$ m/s. The simulations of directional solidification [16] demonstrate that the interface remains stable for velocities above 0.024 m/s. Hence, qualitatively, the growth behaviour of the dendrite is consistent with the significant solute trapping.

It should be concluded that a crystal morphology depending on the heat extraction rate varies from smooth primary stalks without secondary arms (low cooling rate) to well-developed dendritic structure to planar or cellular shape at extremely high cooling rate.

6. CONCLUSIONS

The presented results are thought to be a first attempt to model non-isothermal dendritic solidification of a binary alloy. Removing the isothermal assumption makes the computations much more time consuming, due to the large difference between thermal and mass diffusivities.

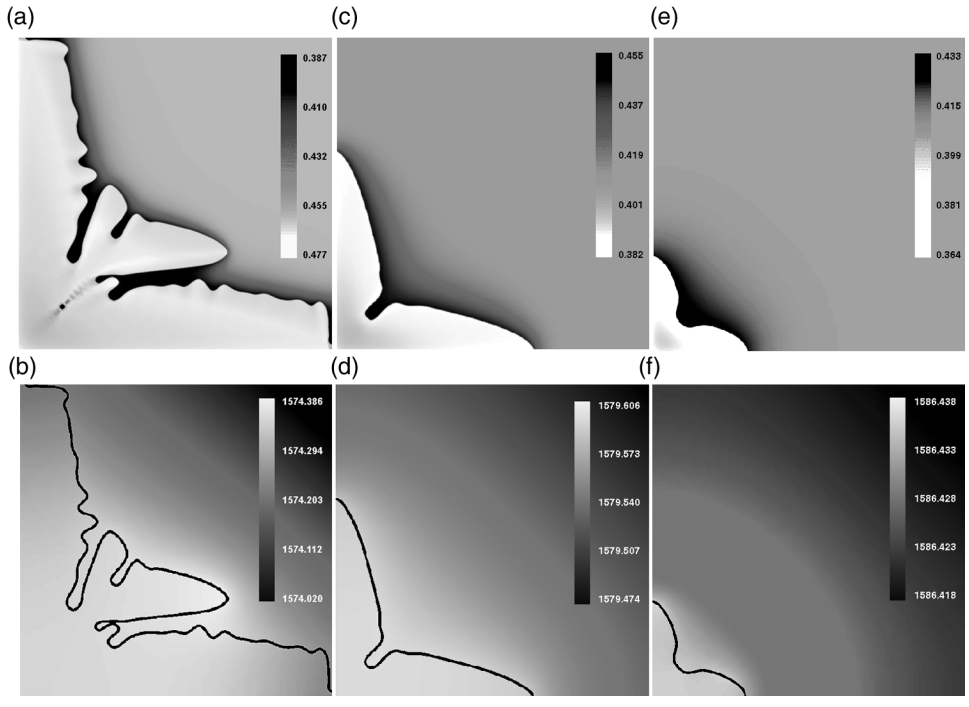


Fig. 7. Solute and temperature redistributions (top and bottom rows, respectively) obtained in the "small" box (2.28×10^{-5} m), for the cooling rates 1.3×10^{5}, 3.4×10^{4} and 2.1×10^{3} K/s varying from the left column to the right one. Time is 1.4 ms. The concentration and the temperature fields employ the same color scheme as in Figs. 2 and 3, respectively. The black line shown with the temperature field represents the location of the solid/liquid interface.

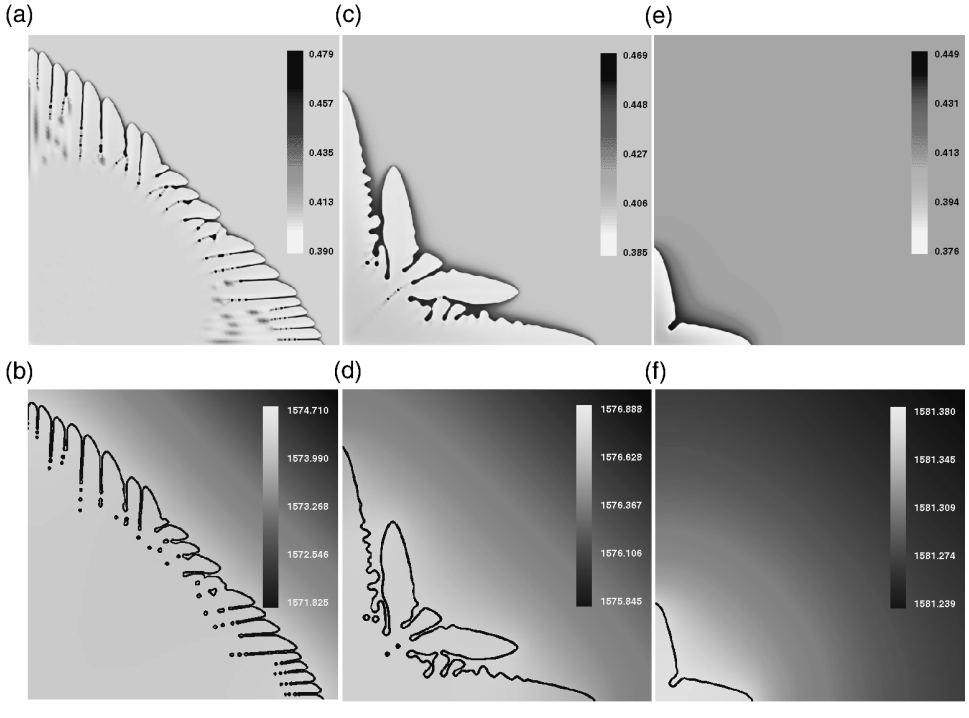


Fig. 8. Solute and temperature redistributions (top and bottom rows, respectively) obtained in the "large" box (6.9×10^{-5} m), for the cooling rates 1.3×10^{5}, 3.4×10^{4} and 2.1×10^{3} K/s varying from the left column to the right one. (a) and (b) are given at time 1.7 ms, while the others are at 2.5 ms.

For low cooling rate and many nuclei the spatial temperature variation is small and may be neglected, and thus the isothermal approach is applicable. However, the non-isothermal effect becomes visible for higher cooling rate and fewer nuclei, when the spatial temperature difference is not small compared to the difference between solidus and liquidus in the phase diagram.

On increasing the cooling rate, the growth eventually becomes governed by thermal diffusion rather than redistribution of solute. Due to limitations on the width of the diffuse interface the results for the highest cooling rate, showing strong solute trapping, may not be quantitatively correct. However, qualitatively the predicted behaviour is in agreement with what is expected at high cooling rates.

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